

Soldering/De-soldering Stations With a Pool of New Features

The growth of the printed circuit board assembly industry has increased the demand for soldering and de-soldering equipment, but selecting the right soldering station is still a concern

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Selecting the right soldering/de-soldering (S/D) station is very important for any project when it comes to printed circuit boards (PCBs) and semiconductors. Occasionally, we end up buying a cheap, unreliable S/D station that either causes damage to components or stops working after the first or second use. The hazards of selecting an incorrect S/D station were discussed in the buyers' guide on re-work stations published in January 2014 issue of *Electronics For You*.

It is advisable to know the required parameters before selecting an S/D station. But to know the parameters, it is essential that we know the basic requirement of the project in hand. From electrical to design parameters of the S/D, all should comply with the project to enable an efficient and safe working environment.

The next thing you should know is the new styles of S/D stations available. There could be S/D station styles with many old features incorporated along with the new ones. Information about the new styles of S/D will not only increase your options but also help you identify features you should look for.

Selection parameters

You must be aware of the common features that are considered as selection parameters for an S/D station, such as a replaceable iron tip, temperature control setting, pre-set temperature option, lightweight iron rod and output voltage of the iron. Apart from these, there are other important parameters that we generally miss out. These are electrostatic discharge (ESD)-safe de-



AT8586 thermo-control anti-static soldering station

What is an S/D station?

You must be aware of the soldering iron that you use in your daily life—iron rod with a plug-in power cord. Compared to this, a soldering or a de-soldering station is totally different. An S/D station has a soldering iron or a de-soldering gun/iron/tweezers or both, wipe sponge and handle to hold the iron, along with temperature settings to control the working of the S/D. A soldering station can be considered as the modern form of a simple soldering iron, which is designed to use an electronic solder (a combination of rosin and electronic-grade flux). The electronic-grade solder prevents damage to the electronic components.

An S/D station is more of a professional soldering system with a small-sized soldering iron that is usually lighter than the plug-in type soldering iron, and a replaceable tip with the required shape and size.

sign, thermal recovery, temperature accuracy and tip-to-ground ratings.

ESD-safe designs. All S/D stations work on electricity that gets converted into heat, creating an electric field around these. This may lead to electrostatic discharge at times, further creating device failures during S/D. The devices that we work on nowadays are pretty compact and delicate to escape the damage caused. Thus, it is essential to look for an ESD-safe soldering station that prevents static from building up by grounding the iron.

Thermal recovery. The thermal

recovery of an S/D is directly proportional to its power. An S/D station with low power will lose heat fast and will have a lower thermal recovery. Thus, an S/D station with low power will be slower to work with. Since thermal recovery of any S/D station is not usually mentioned on it, you should look for its power ratings. An S/D with a power rating of 50W to 80W is considered to be the best for fast operation, whereas an S/D station with power rating of 20W to 30W, or even lower, is unsuitable for most projects.

Temperature accuracy. It is natural